↗ Terminology

Alphabet (Σ) : The 'Character Class'

↳ A set of symbols

$\Sigma = \{0, 1\}$: binary alphabet

$\Sigma = \{0,1,2,3,4,5,6,7\}$: octal alphabet

$\Sigma = \{a, b\}$

↳ a, ab, ba, abb

<u>String</u> : Sentence or Word

↳ Sequence of symbols

↳ Finite in length

$S = abbadc$ length of $s = |s|$

Empty String (ε)

↳ a string where $|\varepsilon| = 0$

<u>Language</u>: A set of ~~language~~ strings over some fixed alphabet

Example:

$L1 = \{a, baa, bccb\}$

$L2 = \{\}$ ← They are different

$L3 = \{\varepsilon\}$ ←

$L4 = \{\varepsilon, ab, abab, ababab, \ldots\}$

↑ each string is finite in length but set may contain infinite elements mShayan

↗ Terminology

Prefix of string

↳ String obtained by removing zero or more trailing symbols

$S = $ Hello

Prefixes : ε, H, He, Hel, Hell, Hello

Suffix of string

↳ String obtained by removing zero or more leading symbols

$S = $ Hello

Suffixes : Hello, Hell, Hel, He, H, ε

Proper Suffix & Proper Prefix :

$$PP = S[0] \neq \varepsilon$$

$$PS = S[n-1] \neq \varepsilon$$

Subsequence :

A lo$\boxed{\text{ng sen}}$tence .

sub sequence

Kleene Closure

a, bc

a, bc

aa, abc, abc, bcbc

Let $L = \{a, bc\}$

$L^0 = \{\varepsilon\}$

$L^1 = \{a, bc\}$

$L^2 = LL = \{aa, abc, bca, bcbc\}$

$L^3 = LLL = \{aaa, aabc, abca, abcbc, bcaa, bcabc, bcbca, bcbcbc\}$

$L^n = L^{n-1}L = LL^{n-1}$

$\therefore$ Kleene Closure = 0 or more.

$$L^* = \{\underbrace{\varepsilon}_{L^0}, \underbrace{a, bc, aa, abc, bca, bcbc}_{L^1}, \underbrace{\ldots\ldots}_{L^2}\}$$

Positive Closure

1 or more

$L^+ = L^*$ BUT 'ε' not included

Regular Expressions

$a|b|cab = \{a, b, cab\}$.

$b^* = \{\varepsilon, b, bb, bbb, \ldots\}$ //0 or more

$a|b^* = \{a, \varepsilon, b, bb, bbb, \ldots\}$

$(a|b)^* = \{\varepsilon, a, b, aa, ab, ba, bb, aaa, \ldots\}$

set of all strings of a's and b's including 'ε'

$L(a|b)^* = L(a^*) \cup L(b^*)$

$= \varepsilon, a, aa, aaa$

ε, b, bb, bbb

Algebric laws of Regular Expressions

Failure Function

Q| The Fibbonacci Strings

$S1 = b$

$S2 = a$

$S_n = S_{n-1} S_{n-2} \quad n>2$

Construct failure function for S6.

6) Shayan

$S_3 = S_2 S_1 = ab$

$S_4 = S_3 S_2 = aba$

$S_5 = S_4 S_3 = abaab$

$S_6 = S_5 S_4 = \underbrace{abaab}_{S_5}\underbrace{aba}_{S_4}$

$\therefore \boxed{S_6 = abaababa}$

String size	Proper Prefix	Proper Suffix
$\lvert S\rvert = 1$ $S = a$	—	—
$\lvert S\rvert = 2$ $S = ab$	a	b
$\lvert S\rvert = 3$ $S = aba$	$\boxed{a} = 1$ ab	$\boxed{a}$ 1 ba
$\lvert S\rvert = 4$ $S = abaa$	$\boxed{a} = 1$ ab aba	$\boxed{a} = 1$ aa baa
$\lvert S\rvert = 5$ $S = abaab$	a $\boxed{ab} = 2$ aba abaa	b $\boxed{ab} = 2$ aab baab
$\lvert S\rvert = 6$ $S = abaaba$	a ab $\boxed{aba} = 3$ abaa abaab	a ba $\boxed{aba} = 3$ aaba baaba

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$\lvert s \rvert = 7$ s= abaabab	a $\boxed{ab}$ aba abaa abaab abaaba	abab aabab baabab
$\lvert s \rvert = 8$ s= abaababa	a ab $\boxed{aba}$ abaa abaab abaaba abaabab	a ba $\boxed{aba}$ baba ababa aababa baababa

S	1	2	3	4	5	6	7	8
f(s)	0	0	1	1	2	3	2	3

Context-Free Grammar

A grammar / formal system used to describe a class of languages,

Purpose of CFG :

→ To list all strings in a language using a set of rules (production rules)

→ Extends the capabilities of Regular Expressions (RE) & Finite Automata.

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A CFG is defined by—

1) Non-terminal Symbols/Variables:

↳ Abstract Category (e.g. E, A...)

2) Terminal Symbols/Alphabet:

↳ Actual characters/Tokens,

↳ id, +, -, a, b, ...

3) Production Rules:

Specify how non-terminals can be replaced with other non-terminals/terminals.

↳ $\underline{E} \rightarrow \underline{E} + \underline{E}$,

$\underline{E} \rightarrow id$ ...

4) Starting Symbol:

The starting non-terminal, which the grammar is first defined.